

Trainings ▾

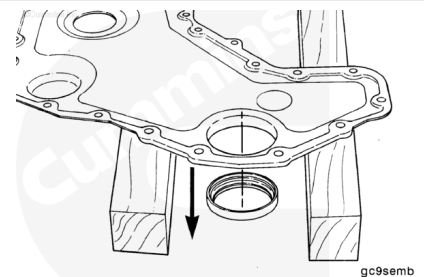
Remove

Front Gear Train

Note : For front gear train engines, the front gear cover must be removed in order to remove the front crankshaft seal. See the next disassembly step for removing the front gear cover.

Support the front gear cover on a flat work surface with wooden blocks. Using a suitable punch and hammer, drive the old seal out of the front gear cover from the back side of the cover to the front side.

Note : Some engines may have an addition dust seal installed in front of the front crankshaft seal.



Rear Gear Train

Use tool, Part Number 3164659, to remove the front crankshaft seal from the front gear cover.

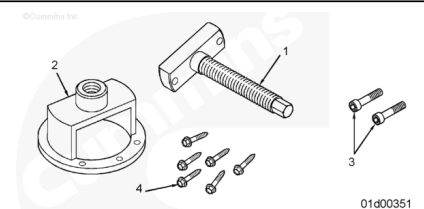


Table 1.
Front
Crankshaft
Seal
Replacer Kit,
Part Number
3164659

Item Number	Part Number	Description	Quantity
1	3164667	Replace screw assembly	1
2	3164661	Crankshaft seal replacer	1
3	3164239	Socket head capscrew, M12 x 1.25 x 60 mm	2

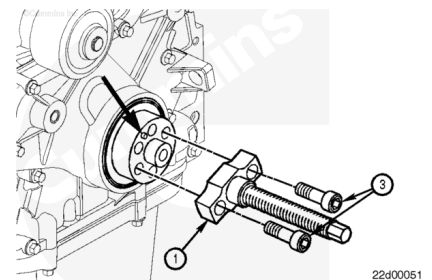
Table 1.
Front
Crankshaft
Seal
Replacer Kit,
Part Number
3164659

Item Number	Part Number	Description	Quantity
4	3164217	Sheet metal screw, Number 10 x 25.4 mm [1 in] long	25 (6 shown)
Not shown	3164218	Drill. 3.57 mm [9/64 in]	1

Note : The front gear cover does not need to be removed to remove and install the crankshaft seal.

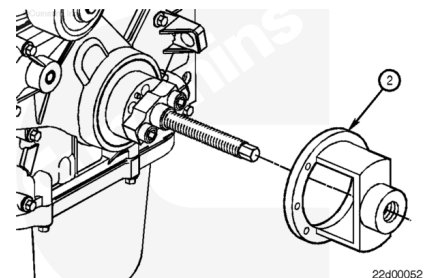
Mount the replacer screw assembly (1) onto the crankshaft nose.

Install the two M12 x 1.25 x 60-mm socket head capscrews (3).



Lubricate the replacer screw with anti-seize or a suitable grease.

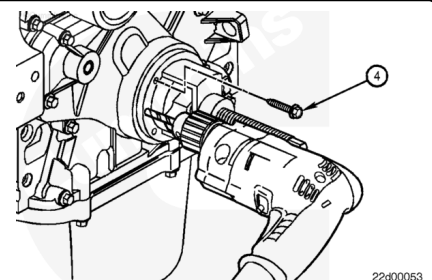
Hold the replacer screw and install the crankshaft seal replacer (2) onto the replacer screw assembly. Advance the crankshaft seal replacer toward the seal by rotating it **clockwise** until it is positioned against the front gear cover.



⚠ CAUTION ⚠

Drill the hole carefully and straight to reduce the possibility of damage to the front cover or the crankshaft.

Service Tip: Because of space restrictions, it may be necessary to use a compact right angle drill. Also, it may be necessary to shorten the drill bit used to drill the sheet metal



screw holes.

Note : The flange of the crankshaft seal replacer is 8 mm [0.32 in] thick.

Mark the drill to a depth of 18 mm [0.71 in] with tape for drill depth control and apply grease to the drill to catch the chips. Stop frequently to remove the chips.

With the crankshaft seal replacer positioned against the front gear cover, drill one hole 10 mm [0.39 in] deep. Make sure the marking tape has **not** moved from the original position.

Install one sheet metal screw (4) into the seal to hold the crankshaft seal replacer in position.

Drill five additional holes 10 mm [0.39 in] deep and install the additional five sheet metal screws.

Note : Make sure all six sheet metal screws are threaded through both the inner and outer seal casings. The sheet metal screws need to be uniformly tightened in order to pull out the inner and outer casings of the seal together.

⚠ CAUTION ⚠

Do not use an impact wrench or air tools. Doing so can damage the tool.

Slowly rotate the replacer screw **clockwise** until the seal is removed. Do **not** exceed torque value.

Torque Value: 33 n·m [45 ft-lb]

Note : If the sheet metal screws pull out of the seal or **only** the inner casing pulls out, stop the removal operation. Rotate the replacer screw **counterclockwise** to force the inner casing back. Remove the sheet metal screws, slightly reorient the tool, drill new holes, and install the sheet metal screws in the new locations.

Complete the removal procedure, remove the tool, and discard the old seal.

